

MTE 544 Assignment 1

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1 Question 1: Mobile Manipulator

(a) The transformation matrix between two reference frames is defined as:

$$\mathcal{G}_b^a = \begin{bmatrix} R_b^a & q \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & x_b \\ \sin \theta & \cos \theta & y_b \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

where $\bar{p}^a = \mathcal{G}_b^a \bar{p}^b$ with homogeneous coordinates $\bar{p} = \begin{bmatrix} p \\ 1 \end{bmatrix}$.

Starting from the end-effector frame $\{E\}$ and proceeding back to the global spatial frame $\{s\}$:

1. Transformation from $\{E\}$ to $\{2\}$ ($\mathcal{G}_E^2$): The end-effector frame $\{E\}$ has its origin displaced by link length l_2 along the X_2 axis. The rotation is fixed at $\psi = -\frac{\pi}{2}$:

$$\mathcal{G}_E^2 = \begin{bmatrix} \cos \psi & -\sin \psi & l_2 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\psi = -\frac{\pi}{2}} \mathcal{G}_E^2 = \begin{bmatrix} 0 & 1 & l_2 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

2. Transformation from $\{2\}$ to $\{1\}$ ($\mathcal{G}_2^1$): Reference frame $\{2\}$ is displaced by link length l_1 along the X_1 axis with joint rotation γ :

$$\mathcal{G}_2^1 = \begin{bmatrix} \cos \gamma & -\sin \gamma & l_1 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3)$$

3. Transformation from $\{1\}$ to $\{B\}$ ($\mathcal{G}_1^B$): Reference frame $\{1\}$ is displaced by distance l_B along the X_B axis with joint rotation α :

$$\mathcal{G}_1^B = \begin{bmatrix} \cos \alpha & -\sin \alpha & l_B \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (4)$$

4. Transformation from $\{B\}$ to $\{s\}$ ($\mathcal{G}_B^s$): The base frame $\{B\}$ has its origin at (x, y) in the global reference frame with heading angle θ :

$$\mathcal{G}_B^s = \begin{bmatrix} \cos \theta & -\sin \theta & x \\ \sin \theta & \cos \theta & y \\ 0 & 0 & 1 \end{bmatrix} \quad (5)$$

5. Overall Transformation from $\{E\}$ to $\{s\}$ ($\mathcal{G}_E^s$): The overall transformation from frame $\{E\}$ to the global frame $\{s\}$ is:

$$\mathcal{G}_E^s = \mathcal{G}_B^s \mathcal{G}_1^B \mathcal{G}_2^1 \mathcal{G}_E^2 \quad (6)$$

Multiplying and simplifying yields:

$$\mathcal{G}_E^s = \begin{bmatrix} \sin(\theta + \alpha + \gamma) & \cos(\theta + \alpha + \gamma) & x + l_B \cos \theta + l_1 \cos(\theta + \alpha) + l_2 \cos(\theta + \alpha + \gamma) \\ -\cos(\theta + \alpha + \gamma) & \sin(\theta + \alpha + \gamma) & y + l_B \sin \theta + l_1 \sin(\theta + \alpha) + l_2 \sin(\theta + \alpha + \gamma) \\ 0 & 0 & 1 \end{bmatrix} \quad (7)$$

(b) Given the numerical configuration:

$$\theta = -\frac{\pi}{2}, \quad \alpha = \frac{\pi}{6}, \quad \gamma = \frac{\pi}{4}, \quad l_1 = 1, \quad l_2 = 1, \quad l_B = 0.5, \quad x = 0.8, \quad y = 0.8$$

Substituting these values into the overall transformation matrix yields:

$$\mathcal{G}_E^s = \begin{bmatrix} -0.2588 & 0.9659 & 2.2659 \\ -0.9659 & -0.2588 & -0.8248 \\ 0 & 0 & 1.0000 \end{bmatrix} \quad (8)$$

To transform the given points $p_1^E = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $p_2^E = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$ to the spatial frame $\{s\}$, we express them in homogeneous coordinates $\bar{p}_i^E = \begin{bmatrix} p_i^E \\ 1 \end{bmatrix}$ and apply $\bar{p}_i^s = \mathcal{G}_E^s \bar{p}_i^E$:

$$\bar{p}_1^s = \mathcal{G}_E^s \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2.2659 \\ -0.8248 \\ 1.0000 \end{bmatrix} \implies p_1^s = \begin{bmatrix} 2.2659 \\ -0.8248 \end{bmatrix} \quad (9)$$

$$\bar{p}_2^s = \mathcal{G}_E^s \begin{bmatrix} 0.5 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3.1024 \\ -1.5666 \\ 1.0000 \end{bmatrix} \implies p_2^s = \begin{bmatrix} 3.1024 \\ -1.5666 \end{bmatrix} \quad (10)$$

(c) A target object point is measured in the robot base frame $\{B\}$ as $q^B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, with homogeneous

coordinates $\bar{q}^B = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

The kinematic transformation from the end-effector frame $\{E\}$ to the base frame $\{B\}$ is given by the chain of transformations as we have seen before in part (a):

$$\bar{q}^B = \mathcal{G}_1^B \mathcal{G}_2^1 \mathcal{G}_E^2 \bar{q}^E = \mathcal{G}_E^B \bar{q}^E \quad (11)$$

To express the target point in the end-effector frame $\{E\}$ for grasping, we invert this transformation. Note this means inverting the matrices as well as the order of multiplication:

$$\bar{q}^E = (\mathcal{G}_E^B)^{-1} \bar{q}^B = (\mathcal{G}_E^2)^{-1} (\mathcal{G}_2^1)^{-1} (\mathcal{G}_1^B)^{-1} \bar{q}^B \quad (12)$$

Evaluating the transformation with the numerical configuration from part (b):

$$\bar{q}^E = (\mathcal{G}_E^2)^{-1}(\mathcal{G}_2^1)^{-1}(\mathcal{G}_1^B)^{-1} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.7418 \\ 0.3542 \\ 1.0000 \end{bmatrix} \quad (13)$$

Thus the location of the point in the end-effector frame $\{E\}$ is:

$$q^E = \begin{bmatrix} -0.7418 \\ 0.3542 \end{bmatrix} \quad (14)$$

2 Question 2: Differential-Drive Simulation

Simulation Approach

We want to find the pose variables $x(t), y(t), \theta(t)$. At each time step, we increment the current position as such:

$$p_{\text{new}} = p_{\text{old}} + \dot{p} \Delta t \quad (15)$$

where,

$$\dot{p} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix} \quad (16)$$

- (a) We simulate the differential-drive robot for a duration of 30s with a timestep of $\Delta t = 0.1$ s. The resulting 2D trajectories (x vs y) and state variables (x, y, θ vs time) for each case are shown below:

i. Constant Velocities

Case 1: $[v, \omega]^\top = [1, 0]^\top$

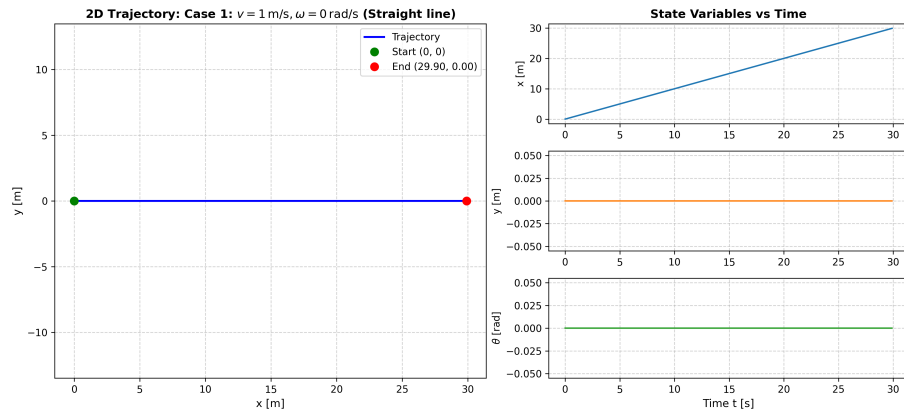


Figure 1: Trajectory and state variables for Case 1 ($v = 1$ m/s, $\omega = 0$ rad/s).

Observations: With zero angular velocity ($\omega = 0$), the robot's heading remains constant at $\theta(t) = 0$ rad. The linear velocity drives the robot in a straight line along the global X -axis, so we see only x increase, and y and θ remain constant.

Case 2: $[v, \omega]^\top = [0, 0.5]^\top$

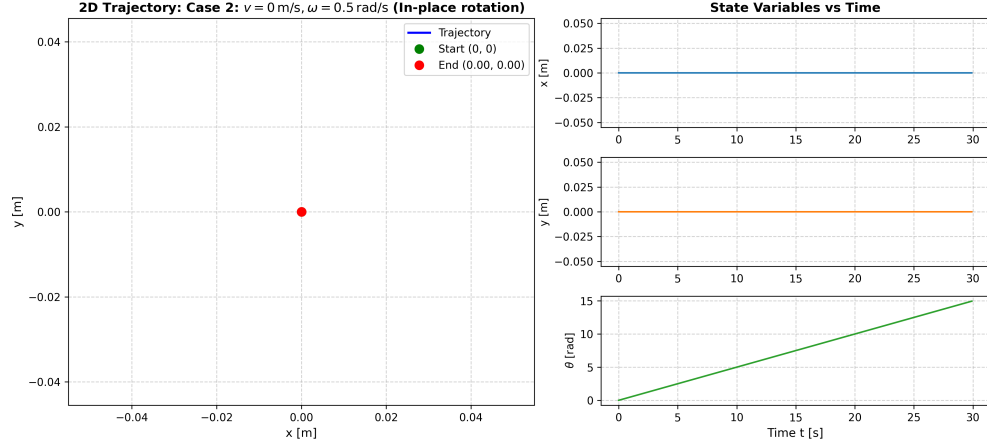


Figure 2: Trajectory and state variables for Case 2 ($v = 0 \text{ m/s}, \omega = 0.5 \text{ rad/s}$).

Observations: With zero linear velocity ($v = 0$), there is no translation, and the robot rotates strictly in place about its center. The position coordinates don't change. The heading $\theta(t)$ increases linearly.

Case 3: $[v, \omega]^\top = [1, 0.5]^\top$

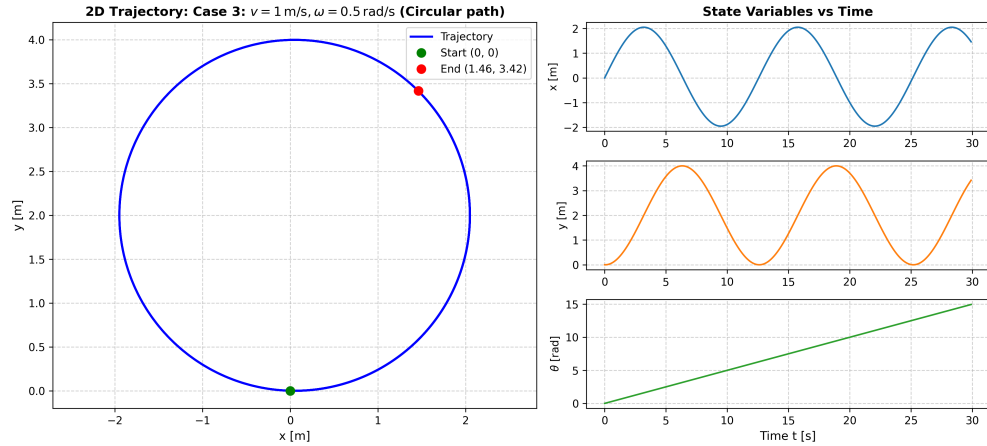


Figure 3: Trajectory and state variables for Case 3 ($v = 1 \text{ m/s}, \omega = 0.5 \text{ rad/s}$).

Observations: Combining constant linear and angular velocities yields a circle. The robot traces a circle of radius 2m centered at (0, 2) m. The position states $x(t)$ and $y(t)$ oscillate sinusoidally with an amplitude of 2m which is a neat correlation to notice since indeed as we trace around a circle we expect the x and y components to form cos and sin respectively.

ii. Velocity Profiles

Profile:
$$\begin{bmatrix} v(t) \\ \omega(t) \end{bmatrix} = \begin{bmatrix} 1 + 0.2 \sin(t) \\ 0.2 + 0.6 \cos(t) \end{bmatrix}$$

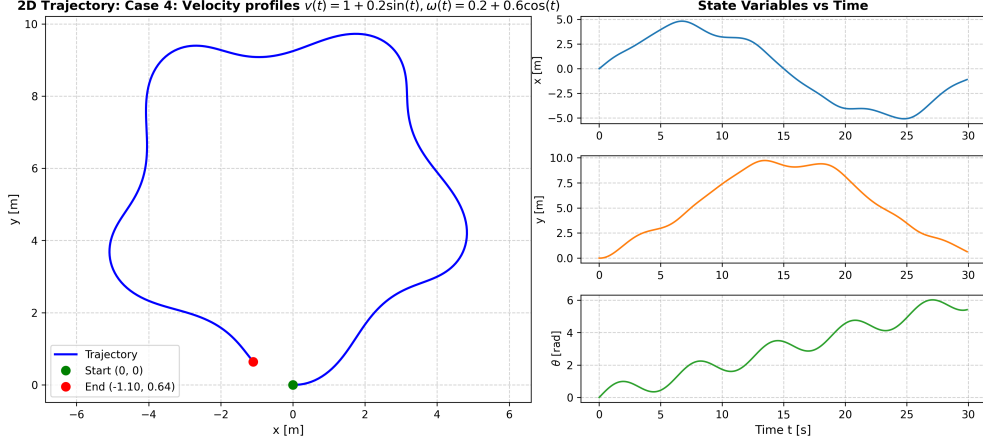


Figure 4: Trajectory and state variables for Case 4 (time-varying velocity profiles).

Observations: The angular rate $\omega(t)$ has a "bias" of 0.2 rad/s, so it produces an overall net counter-clockwise trajectory while the $0.6 \cos(t)$ component periodically wiggles the curvature. Coupled with the oscillating speed $v(t)$, the robot traces an undulating, complex, yet periodic trajectory.

- (b) Given robot parameters $T = 0.3$ m (track width) and $r = 0.15$ m (wheel radius).

The linear velocities at the left and right wheels are:

$$v_L^w = \omega R_L = \omega \left(R - \frac{T}{2} \right) = v - \frac{T\omega}{2} \quad (17)$$

$$v_R^w = \omega R_R = \omega \left(R + \frac{T}{2} \right) = v + \frac{T\omega}{2} \quad (18)$$

The corresponding wheel speeds $u_r(v, \omega)$ and $u_l(v, \omega)$ (in rad/s) are:

$$u_r(v, \omega) = \frac{1}{r} v_R^w = \frac{1}{r} \left(v + \frac{T\omega}{2} \right) = \frac{1}{0.15} (v + 0.15\omega) \quad (19)$$

$$u_l(v, \omega) = \frac{1}{r} v_L^w = \frac{1}{r} \left(v - \frac{T\omega}{2} \right) = \frac{1}{0.15} (v - 0.15\omega) \quad (20)$$

Applying this to the three constant velocity cases:

1. For $[v, \omega]^\top = [1, 0]^\top$:

$$u_r(v, \omega) = 6.667 \text{ rad/s}, \quad u_l(v, \omega) = 6.667 \text{ rad/s}$$

2. For $[v, \omega]^\top = [0, 0.5]^\top$:

$$u_r(v, \omega) = 0.5 \text{ rad/s}, \quad u_l(v, \omega) = -0.5 \text{ rad/s}$$

3. For $[v, \omega]^\top = [1, 0.5]^\top$:

$$u_r(v, \omega) = 7.167 \text{ rad/s}, \quad u_l(v, \omega) = 6.167 \text{ rad/s}$$

Simulation Modification for Wheel Speed Inputs: Solving the system of linear equations in (19) and (20):

$$\omega = r \frac{u_r - u_l}{T}, \quad v = r \frac{u_r + u_l}{2} \quad (21)$$

Substituting these directly into the kinematic state space equation from earlier:

$$\begin{aligned} \dot{p} &= \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{r}{2}(u_r + u_l) \\ \frac{r}{T}(u_r - u_l) \end{bmatrix} \\ &= \begin{bmatrix} \frac{r}{2} \cos \theta (u_r + u_l) \\ \frac{r}{2} \sin \theta (u_r + u_l) \\ \frac{r}{T} (u_r - u_l) \end{bmatrix} \\ &= \begin{bmatrix} \frac{r}{2} \cos \theta & \frac{r}{2} \cos \theta \\ \frac{r}{2} \sin \theta & \frac{r}{2} \sin \theta \\ \frac{r}{T} & -\frac{r}{T} \end{bmatrix} \begin{bmatrix} u_r \\ u_l \end{bmatrix} \end{aligned} \quad (22)$$

This matrix expression for $\dot{p}$ in terms of $[u_r, u_l]^\top$ now substitutes in exactly where $\dot{p}$ was computed previously in the Forward Euler update loop ($p_{\text{new}} = p_{\text{old}} + \dot{p} \Delta t$).

3 Question 3: 3-Omnidirectional Simulation

- (a) To derive the equations of motion for the 3-wheel omnidirectional robot, we define the coordinate frames and geometric parameters as shown in Figure 5.

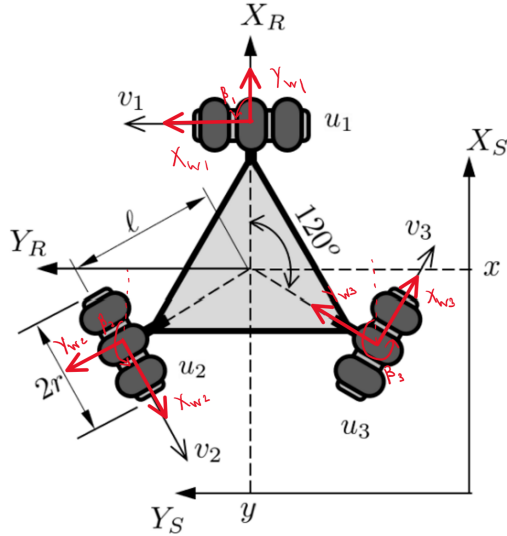


Figure 5: 3-Omnidirectional robot frame assignments and wheel configuration.

From the lecture derivation for omnidirectional wheeled mobile bases, the rotational speed u_i of wheel i is related to the robot velocity $\dot{q} = [\dot{x} \ \dot{y} \ \dot{\theta}]^\top$ by:

$$u_i = \frac{1}{r_i} [1 \ \tan \gamma_i] g_i(\theta) \dot{q} \quad (23)$$

where $g_i(\theta)$ is

$$g_i(\theta) = \begin{bmatrix} \cos(\theta + \beta_i) & \sin(\theta + \beta_i) & X_i \sin \beta_i - Y_i \cos \beta_i \\ -\sin(\theta + \beta_i) & \cos(\theta + \beta_i) & X_i \cos \beta_i + Y_i \sin \beta_i \end{bmatrix} \quad (24)$$

For our symmetrical 3-omniwheel configuration:

- All three wheels share identical radius: $r = r_1 = r_2 = r_3$.
- The roller angle for standard omniwheels is perpendicular to the wheel plane: $\gamma = \gamma_1 = \gamma_2 = \gamma_3 = 0^\circ$.
- The wheel orientation angles (measured from the positive x-axis of the robot frame to the positive x-axis of the wheel frame, shown in Figure 5) are separated by 120° :

$$\beta_1 = 90^\circ, \quad \beta_2 = 210^\circ, \quad \beta_3 = 330^\circ$$

Since $\tan \gamma_i = \tan(0^\circ) = 0$, only the first row of $g_i(\theta)$ contributes to driving the wheel:

$$\begin{aligned} u_i &= \frac{1}{r} [1 \ 0] g_i(\theta) \dot{q} \\ &= \frac{1}{r} \left(\dot{x} \cos(\theta + \beta_i) + \dot{y} \sin(\theta + \beta_i) + \dot{\theta} (X_i \sin \beta_i - Y_i \cos \beta_i) \right) \end{aligned} \quad (25)$$

Each wheel is mounted at a distance L from the robot center. Let α_i be the angle between the positive x-axis of the robot frame and the line connecting the robot center to the center of wheel, which is to say the angle to Y_i , where $\alpha_i = \beta_i - 90^\circ$. The wheel center coordinates (X_i, Y_i) are:

$$\begin{aligned} X_i &= L \cos \alpha_i = L \cos(\beta_i - 90^\circ) = L \sin \beta_i \\ Y_i &= L \sin \alpha_i = L \sin(\beta_i - 90^\circ) = -L \cos \beta_i \end{aligned}$$

Substituting (X_i, Y_i) into the coefficient of $\dot{\theta}$:

$$\begin{aligned} X_i \sin \beta_i - Y_i \cos \beta_i &= (L \sin \beta_i) \sin \beta_i - (-L \cos \beta_i) \cos \beta_i \\ &= L (\sin^2 \beta_i + \cos^2 \beta_i) \\ &= L \end{aligned} \quad (26)$$

This neatly simplifies to the constant distance L for every wheel, which makes sense intuitively because all wheels are symmetrically located at distance L from the center.

Individual Wheel Equations: Substituting (26) back into (25) gives the general relation:

$$u_i = \frac{1}{r} \left(\dot{x} \cos(\theta + \beta_i) + \dot{y} \sin(\theta + \beta_i) + L\dot{\theta} \right) \quad (27)$$

Plugging in the angles $\beta_1 = 90^\circ$, $\beta_2 = 210^\circ$, $\beta_3 = 330^\circ$ for each wheel:

$$u_1 = \frac{1}{r} \left(\dot{x} \cos(\theta + 90^\circ) + \dot{y} \sin(\theta + 90^\circ) + L\dot{\theta} \right) \quad (28)$$

$$u_2 = \frac{1}{r} \left(\dot{x} \cos(\theta + 210^\circ) + \dot{y} \sin(\theta + 210^\circ) + L\dot{\theta} \right) \quad (29)$$

$$u_3 = \frac{1}{r} \left(\dot{x} \cos(\theta + 330^\circ) + \dot{y} \sin(\theta + 330^\circ) + L\dot{\theta} \right) \quad (30)$$

Collecting all three equations into matrix form yields:

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \frac{1}{r} G \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \frac{1}{r} \begin{bmatrix} \cos(\theta + 90^\circ) & \sin(\theta + 90^\circ) & L \\ \cos(\theta + 210^\circ) & \sin(\theta + 210^\circ) & L \\ \cos(\theta + 330^\circ) & \sin(\theta + 330^\circ) & L \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} \quad (31)$$

(b) To simulate the robot, we need the state derivatives $\dot{q} = [\dot{x} \ \dot{y} \ \dot{\theta}]^\top$ as functions of the wheel speeds $u = [u_1 \ u_2 \ u_3]^\top$. Inverting the forward kinematic relation $\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \frac{1}{r} G \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$ yields:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = r G^{-1} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \quad (32)$$

Inverting G symbolically in MATLAB gives the inverse kinematic matrix:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \frac{r}{3} \begin{bmatrix} -2 \sin \theta & -2 \cos \left(\theta + \frac{\pi}{6} \right) & 2 \sin \left(\theta + \frac{\pi}{3} \right) \\ 2 \cos \theta & -2 \cos \left(\theta - \frac{\pi}{3} \right) & -2 \cos \left(\theta + \frac{\pi}{3} \right) \\ \frac{1}{L} & \frac{1}{L} & \frac{1}{L} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \quad (33)$$

I. Rotation Inputs: $u_1 = -2.0$ rad/s, $u_2 = 1.0$ rad/s, $u_3 = 1.0$ rad/s.

We simulate the true motion of the robot for 30s with a timestep of $\Delta t = 0.1$ s, using $r = 0.10$ m and $L = 0.25$ m starting from $(x, y, \theta) = (0, 0, 0)$.

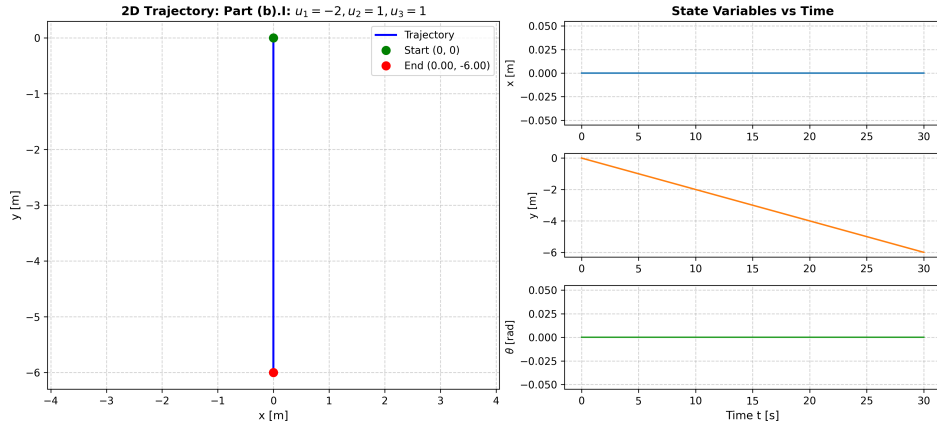


Figure 6: Trajectory and state variables for Part (b).I ($u_1 = -2$ rad/s, $u_2 = 1$ rad/s, $u_3 = 1$ rad/s).

Observations: Since the sum of wheel speeds is $u_1 + u_2 + u_3 = -2 + 1 + 1 = 0$, the net angular velocity $\dot{\theta} = \frac{r}{3L}(u_1 + u_2 + u_3) = 0$ rad/s. The robot doesn't rotate at all, so its heading stays flat at $\theta(t) = 0$ rad. This makes intuitive sense as we can imagine each of the three wheels canceling each other out.

Along the y -axis, all three wheels push together in the $-y$ direction to give a constant speed. As expected, the robot traces a straight line along the $-y$ axis.

II. Designed control signals

1. Straight line with a slope of 60 degrees

To move in a straight line with a slope of 60° , the trajectory must satisfy:

$$\frac{\dot{y}}{\dot{x}} = \tan(60^\circ) = \sqrt{3} \implies \dot{y} = \tan(60^\circ)\dot{x} \quad (34)$$

Choosing a constant linear velocity along the x -axis of $\dot{x} = 1$ m/s yields:

$$\dot{x} = 1 \text{ m/s} \quad (35)$$

$$\dot{y} = \sqrt{3} \text{ m/s} \approx 1.732 \text{ m/s} \quad (36)$$

$$\dot{\theta} = 0 \text{ rad/s} \quad (37)$$

Since the robot starts at orientation $\theta(0) = 0$ and $\dot{\theta} = 0$, the orientation remains $\theta(t) = 0$ for all time. Substituting $\theta = 0$, $\dot{q} = [1, \sqrt{3}, 0]^T$, and wheel radius $r = 0.10$ m into the forward kinematics relation $u = \frac{1}{r}G(0)\dot{q}$:

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} -\sin(0) & \cos(0) & L \\ -\sin(\frac{2\pi}{3}) & \cos(\frac{2\pi}{3}) & L \\ -\sin(\frac{4\pi}{3}) & \cos(\frac{4\pi}{3}) & L \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \frac{1}{0.10} \begin{bmatrix} 0 & 1 & L \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} & L \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} & L \end{bmatrix} \begin{bmatrix} 1 \\ \sqrt{3} \\ 0 \end{bmatrix} \quad (38)$$

Evaluating each wheel velocity:

$$u_1 = \frac{1}{0.10}(0 \cdot 1 + 1 \cdot \sqrt{3} + 0) = \frac{\sqrt{3}}{0.10} = 10\sqrt{3} \approx 17.32 \text{ rad/s} \quad (39)$$

$$u_2 = \frac{1}{0.10} \left(-\frac{\sqrt{3}}{2} \cdot 1 - \frac{1}{2} \cdot \sqrt{3} + 0 \right) = \frac{-\sqrt{3}}{0.10} = -10\sqrt{3} \approx -17.32 \text{ rad/s} \quad (40)$$

$$u_3 = \frac{1}{0.10} \left(\frac{\sqrt{3}}{2} \cdot 1 - \frac{1}{2} \cdot \sqrt{3} + 0 \right) = 0 \text{ rad/s} \quad (41)$$

The simulation is run for $t = 30$ s with initial state $q(0) = [0, 0, 0]^T$.

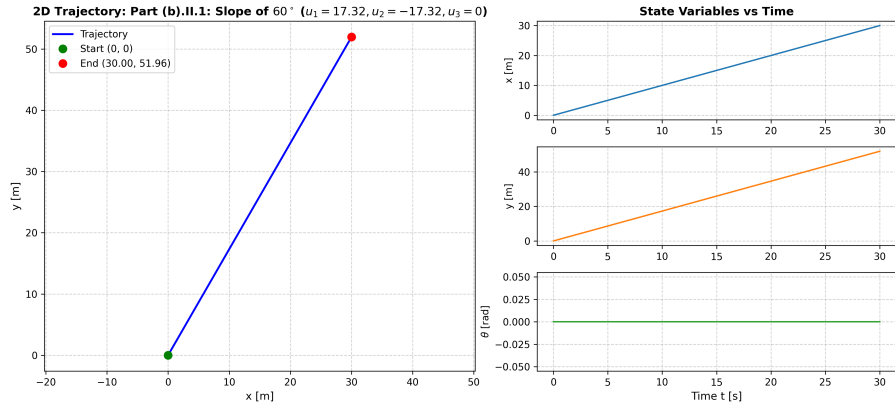


Figure 7: Trajectory and state variables for 60° slope straight line ($u_1 = 17.32 \text{ rad/s}, u_2 = -17.32 \text{ rad/s}, u_3 = 0 \text{ rad/s}$).

Observations: With $u_1 = -u_2$ and $u_3 = 0$, the sum of wheel speeds is $u_1 + u_2 + u_3 = 10\sqrt{3} - 10\sqrt{3} + 0 = 0$, meaning the net rotational speed is exactly zero and the robot maintains $\theta(t) = 0$ throughout. Wheel 3 does not spin at all ($u_3 = 0$) and simply rolls freely on its passive rollers as wheels 1 and 2 drive the robot diagonally. Over 30 seconds, the robot reaches $(x, y) = (30.00, 51.96) \text{ m}$, maintaining the exact commanded slope of $51.96/30.00 = 1.732 = \tan(60^\circ)$.

2. Two-metre diameter circle

To trace a circle with a diameter of $D = 2 \text{ m}$, the path radius is $R = 1 \text{ m}$. As illustrated in Figure 8, we define the instantaneous center of rotation (ICR) such that the robot body frame $\{R\}$ has its forward axis x_R tangent to the trajectory with forward linear speed v , while rotating about the ICR with angular velocity $\omega = \dot{\theta}$.

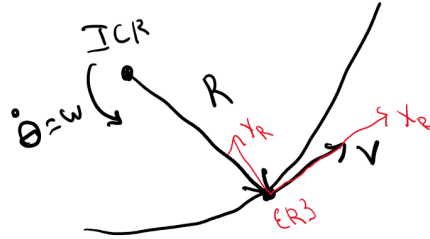


Figure 8: Robot body frame $\{R\}$ aligned tangentially along the circular trajectory of radius R about the ICR.

The relationship between linear velocity and angular velocity for circular motion is:

$$v = \omega R \quad (42)$$

With $R = 1 \text{ m}$, this simplifies to $v = \omega$. Choosing an angular velocity of $\omega = 1 \text{ rad/s} = \dot{\theta}$ yields a linear velocity of $v = 1 \text{ m/s}$.

In the robot body frame $\{R\}$, the velocity components are:

$$\dot{x}^R = v = 1 \text{ m/s} \quad (43)$$

$$\dot{y}^R = 0 \text{ m/s} \quad (44)$$

$$\dot{\theta}^R = \dot{\theta} = 1 \text{ rad/s} \quad (45)$$

To evaluate the forward kinematics, we transform the linear velocity from the robot frame $\{R\}$ to the spatial (inertial) frame $\{S\}$ via rotation matrix $R_R^S(\theta)$:

$$v^S = R_R^S(\theta)v^R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} \quad (46)$$

The spatial state derivative vector is therefore:

$$\dot{q} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 1 \end{bmatrix} \quad (47)$$

Substituting $\dot{q}$ into the forward kinematics relation $u = \frac{1}{r}G(\theta)\dot{q}$:

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} \cos(\theta + 90^\circ) & \sin(\theta + 90^\circ) & L \\ \cos(\theta + 210^\circ) & \sin(\theta + 210^\circ) & L \\ \cos(\theta + 330^\circ) & \sin(\theta + 330^\circ) & L \end{bmatrix} \begin{bmatrix} \cos \theta \\ \sin \theta \\ 1 \end{bmatrix} \quad (48)$$

Using the angle-difference identity $\cos(\theta + \beta_i) \cos \theta + \sin(\theta + \beta_i) \sin \theta = \cos(\beta_i)$, the orientation angle θ cancels out entirely, yielding constant wheel speeds:

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} L \\ L - \frac{\sqrt{3}}{2} \\ L + \frac{\sqrt{3}}{2} \end{bmatrix} \quad (49)$$

Substituting the robot dimensions $r = 0.10$ m and $L = 0.25$ m:

$$u_1 = \frac{0.25}{0.10} = 2.5 \text{ rad/s} \quad (50)$$

$$u_2 = \frac{0.25 - \frac{\sqrt{3}}{2}}{0.10} = 2.5 - 5\sqrt{3} \approx -6.16 \text{ rad/s} \quad (51)$$

$$u_3 = \frac{0.25 + \frac{\sqrt{3}}{2}}{0.10} = 2.5 + 5\sqrt{3} \approx 11.16 \text{ rad/s} \quad (52)$$

The simulation is executed for $t = 30$ s starting from $q(0) = [0, 0, 0]^T$ with timestep $\Delta t = 0.1$ s.

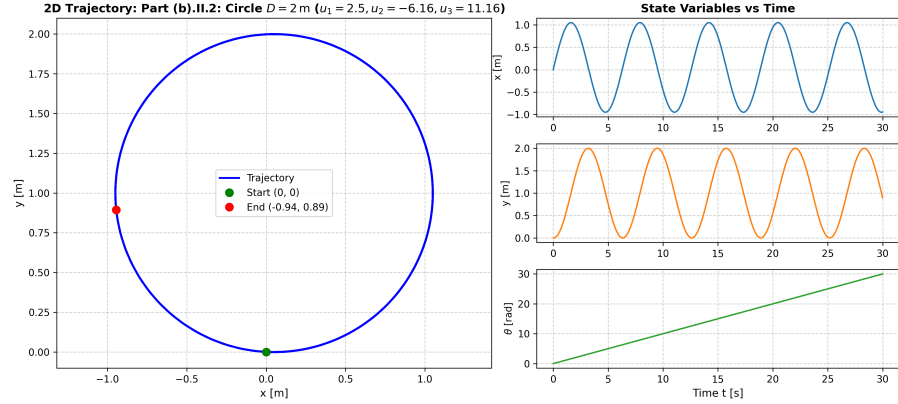


Figure 9: Trajectory and state variables for 2m diameter circle ($u_1 = 2.5$ rad/s, $u_2 = -6.16$ rad/s, $u_3 = 11.16$ rad/s).

Observations: Even though the robot traverses a curved circular path, the required wheel inputs u_1, u_2, u_3 are completely constant! This occurs because the robot's heading $\theta(t)$ rotates at the exact same rate as its tangent velocity vector ($\dot{\theta} = \omega = 1$ rad/s), keeping the relative velocity of the ground in the robot's local frame invariant.

The net rotational velocity is $\dot{\theta} = \frac{r}{3L}(u_1 + u_2 + u_3) = \frac{0.10}{3(0.25)}(2.5 - 6.16 + 11.16) = 1.0$ rad/s, so $\theta(t)$ climbs linearly with a slope of 1 rad/s and completes one full revolution every $2\pi \approx 6.28$ s. In the Cartesian plane, $x(t)$ and $y(t)$ oscillate sinusoidally with amplitude 1.0 m, creating a closed circular path of diameter 2.0 m centered at (0, 1) m. Over the 30-second duration, the robot traces approximately 4.77 full laps around the circle.